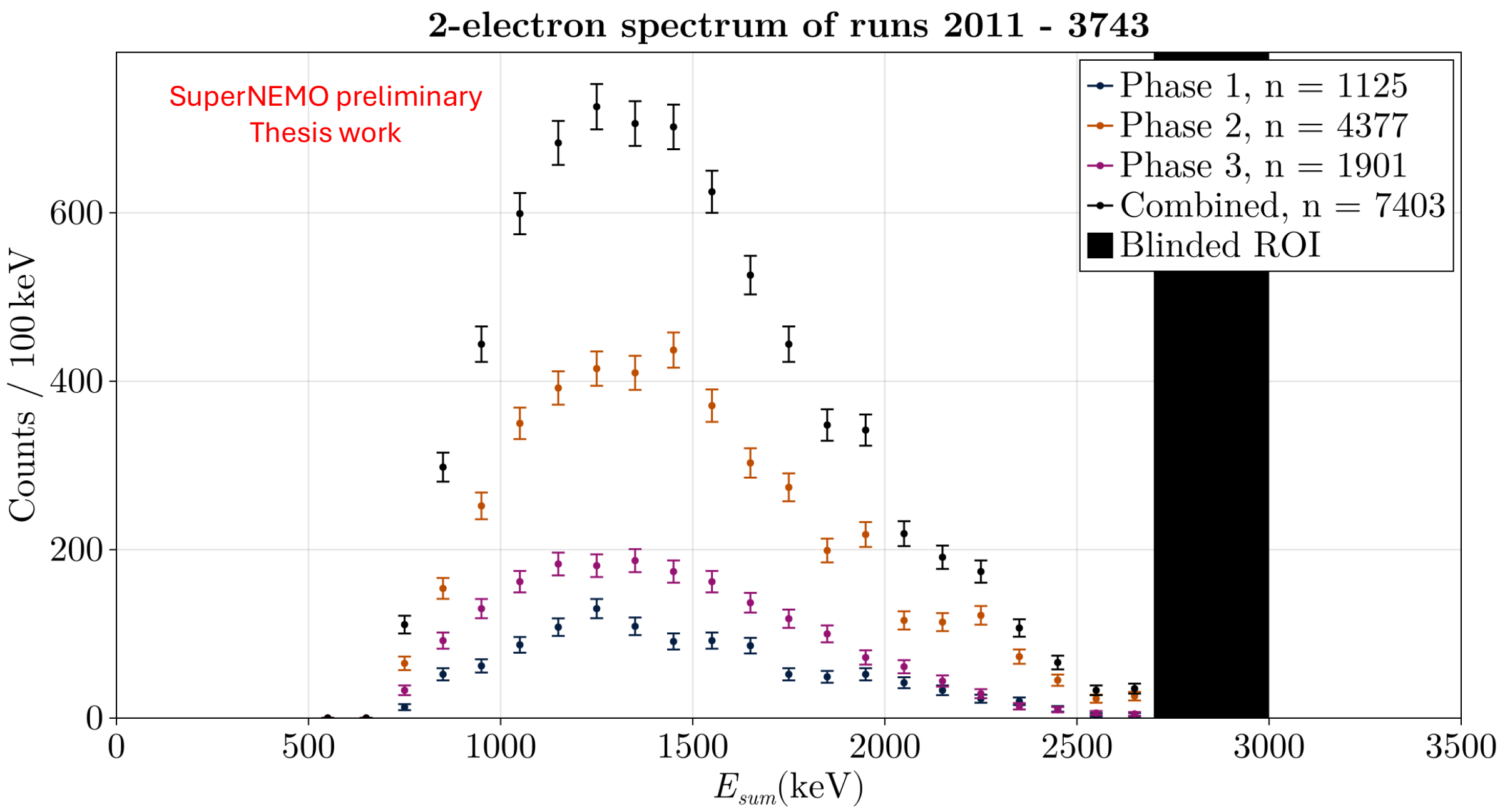
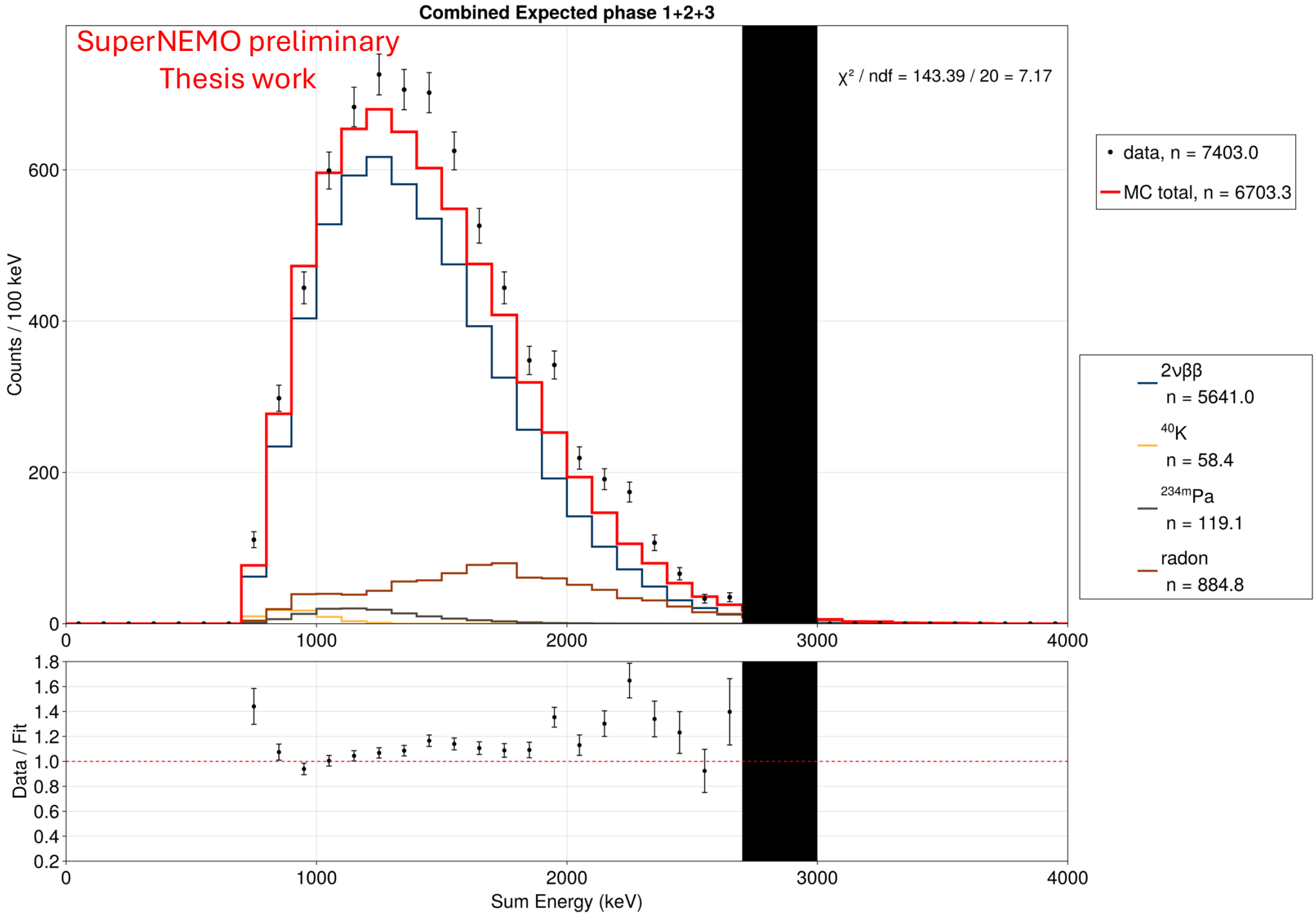
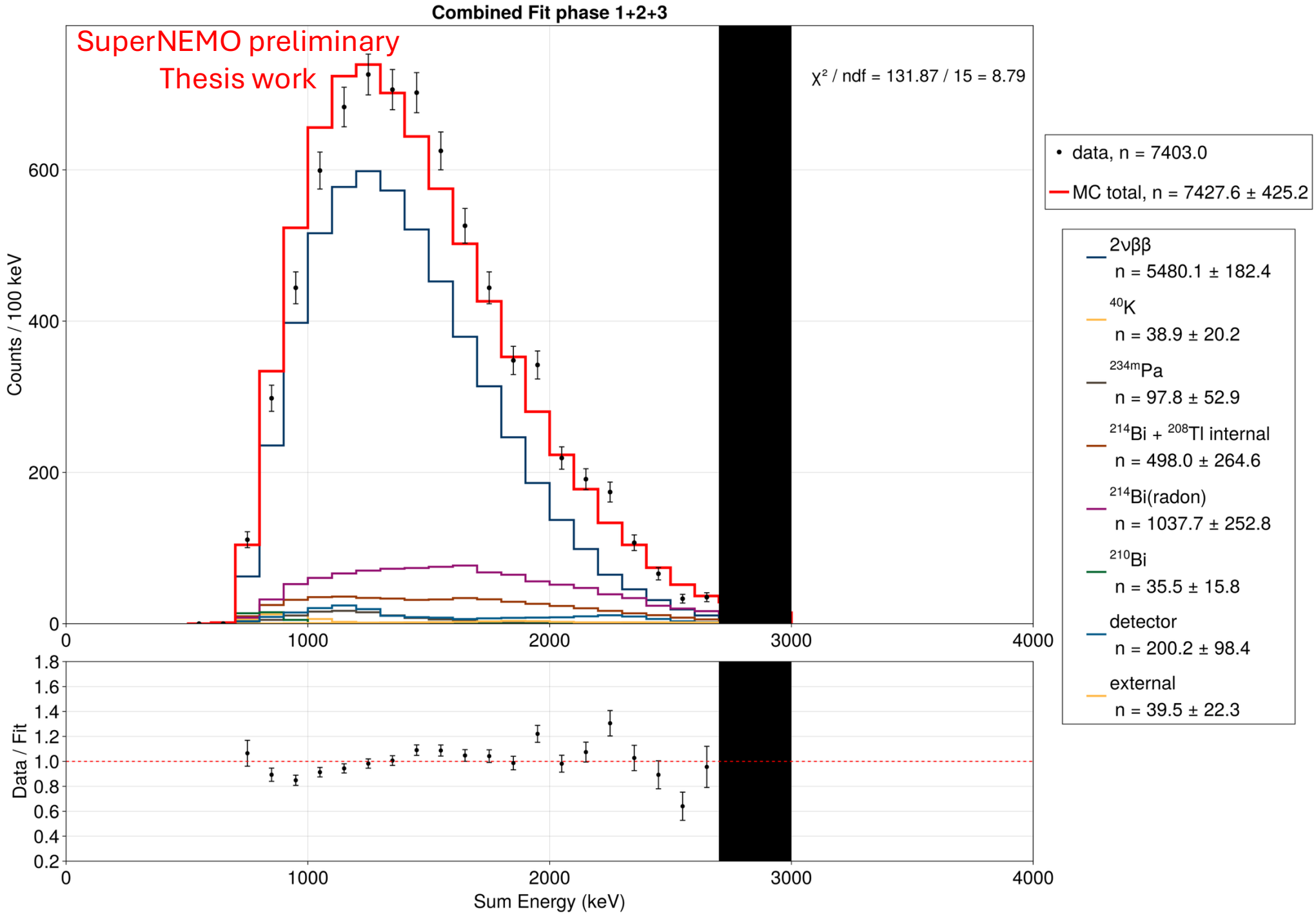
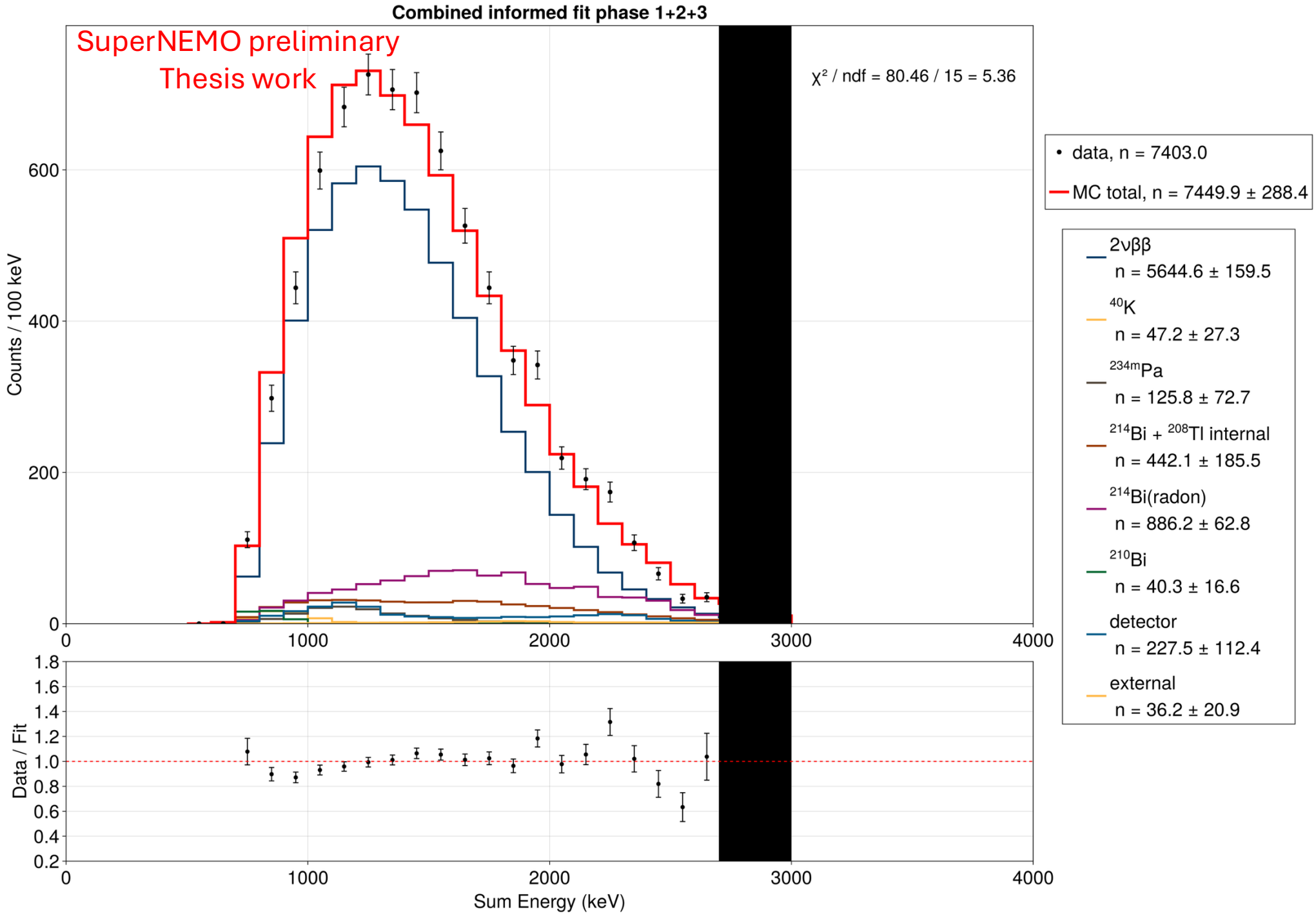

2-electron spectrum of runs 2011 - 3743
SuperNEMO preliminary
Thesis work
Phase 1, n = 1125
Phase 2, n = 4377
Phase 3, n = 1901
Combined, n = 7403
Blinded ROI
Counts / 100 keV
0
200
400
600
0
500
1000
1500
2000
2500
3000
3500
E_{sum}(keV)


Combined Expected phase 1+2+3
SuperNEMO preliminary
Thesis work
χ² / ndf = 143.39 / 20 = 7.17
data, n = 7403.0
MC total, n = 6703.3
2νββ
n = 5641.0
40K
n = 58.4
234mPa
n = 119.1
radon
n = 884.8
Counts / 100 keV
Data / Fit
Sum Energy (keV)


Combined Fit phase 1+2+3
SuperNEMO preliminary
Thesis work
χ² / ndf = 131.87 / 15 = 8.79
data, n = 7403.0
MC total, n = 7427.6 ± 425.2
2νββ
n = 5480.1 ± 182.4
40K
n = 38.9 ± 20.2
234mPa
n = 97.8 ± 52.9
214Bi + 208Tl internal
n = 498.0 ± 264.6
214Bi(radon)
n = 1037.7 ± 252.8
210Bi
n = 35.5 ± 15.8
detector
n = 200.2 ± 98.4
external
n = 39.5 ± 22.3
Counts / 100 keV
Data / Fit
Sum Energy (keV)


Combined informed fit phase 1+2+3
SuperNEMO preliminary
Thesis work
χ² / ndf = 80.46 / 15 = 5.36
data, n = 7403.0
MC total, n = 7449.9 ± 288.4
2νββ
n = 5644.6 ± 159.5
40K
n = 47.2 ± 27.3
234mPa
n = 125.8 ± 72.7
214Bi + 208Tl internal
n = 442.1 ± 185.5
214Bi(radon)
n = 886.2 ± 62.8
210Bi
n = 40.3 ± 16.6
detector
n = 227.5 ± 112.4
external
n = 36.2 ± 20.9
Counts / 100 keV
Data / Fit
Sum Energy (keV)